

Lab 1 : Working with containers

Part 1 :

```
user@deb:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
Digest: sha256:6dc565aa630927052111f823c303948cf83670a3903ffa3849f1488ab517f891
Status: Downloaded newer image for hello-world:latest
```

Hello from Docker!

This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

```
user@deb:~$ docker --version
```

Docker version 28.5.1, build e180ab8

```
user@deb:~$ docker info | grep -i rootless
```

WARNING: No cpuset support

rootless

WARNING: No io.weight support

WARNING: No io.weight (per device) support

WARNING: No io.max (rbps) support

WARNING: No io.max (wbps) support

WARNING: No io.max (riops) support

WARNING: No io.max (wiops) support

```
user@deb:~$ docker container run alpine hostname
```

Unable to find image 'alpine:latest' locally

latest: Pulling from library/alpine

2d35ebdb57d9: Pull complete

Digest: sha256:4b7ce07002c69e8f3d704a9c5d6fd3053be500b7f1c69fc0d80990c2ad8dd412

Status: Downloaded newer image for alpine:latest

b4a075657352

```
user@deb:~$ docker container ls --all
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b4a075657352	alpine	"hostname"	3 minutes ago	Exited (0) 3 minutes ago		admiring_robinson
lafe12c76f4	hello-world	"/hello"	5 minutes ago	Exited (0) 5 minutes ago		condescend_

```
ding_chatterjee
user@deb:~$
```

```

user@deb:~$ docker container run --interactive --tty --rm ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
4b3ffd8ccb52: Pull complete
Digest: sha256:66460d557b25769b102175144d538d88219c077c678a49af4afca6fbfc1b5252
Status: Downloaded newer image for ubuntu:latest
root@3bba2416d2ff:/# ls /
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@3bba2416d2ff:/# ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1  4588  4052 pts/0    Ss   17:57   0:00 bash
root          10  0.0  0.0   7888  4024 pts/0    R+   17:59   0:00 ps aux
root@3bba2416d2ff:/# cat /etc/issue
Ubuntu 24.04.3 LTS \n \l

root@3bba2416d2ff:/# exit
exit
user@deb:~$ cat /etc/issue
Debian GNU/Linux 13 \n \l

user@deb:~$ cat /etc/os-release
PRETTY_NAME="Debian GNU/Linux 13 (trixie)"
NAME="Debian GNU/Linux"
VERSION_ID="13"
VERSION="13 (trixie)"
VERSION_CODENAME=trixie
DEBIAN_VERSION_FULL=13.1
ID=debian
HOME_URL="https://www.debian.org/"
SUPPORT_URL="https://www.debian.org/support"
BUG_REPORT_URL="https://bugs.debian.org/"
user@deb:~$ docker container run \
--detach \
--name mydb \
-e MYSQL_ROOT_PASSWORD=my-secret-pw \
mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
023a182c62a0: Pull complete
f5f78fcd9ccb: Pull complete
494c372d15c3: Pull complete
dcee80f7340c: Pull complete
480d01bd7a6a: Pull complete
834e15e3ed24: Pull complete
c276de9b5571: Pull complete
0cd145fbb449: Pull complete
5a3f7744d0e7: Pull complete
21aa606d8d58: Pull complete
Digest: sha256:569c4128dfa625ac2ac62cdd8af588a3a6a60a049d1a8d8f0fac95880ecdbbe5
Status: Downloaded newer image for mysql:latest
60a403cbe35635b9d449f78e9ac008ba319fa98c720bd8c3518ee60706389818

```

```

user@deb:~$ docker container ls
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
60a403cbe356   mysql:latest   "docker-entrypoint.s..." 6 minutes ago  Up 6 minutes  3306/tcp, 33060/tcp
p   mydb
user@deb:~$ docker container top mydb
UID           PID          PPID          C              STIME          TTY
dnsmasq       5295         5267          3              20:05          ?
00:00:14      mysqld
user@deb:~$ docker exec -it mydb \
mysql --user=root --password=$MYSQL_ROOT_PASSWORD --version
mysql: [Warning] Using a password on the command line interface can be insecure.
mysql Ver 9.5.0 for Linux on x86_64 (MySQL Community Server - GPL)
user@deb:~$ docker exec -it mydb sh
sh-5.1# mysql --user=root --password=$MYSQL_ROOT_PASSWORD --version
mysql: [Warning] Using a password on the command line interface can be insecure.
mysql Ver 9.5.0 for Linux on x86_64 (MySQL Community Server - GPL)

```

Part 2 :

```
app.py requirements.txt Dockerfile X
lab1 > Dockerfile
1 # Use an official Python image

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
docker - lab1 + - [] [] ... | [] X

user@deb:~/Documents$ docker build -t python-app .
y
○ user@deb:~/Documents$ ^C
○ user@deb:~/Documents$ ^C
● user@deb:~/Documents$ ls
lab1
● user@deb:~/Documents$ cd lab1
● user@deb:~/Documents/lab1$ docker build -t python-app .
[+] Building 68.9s (9/9) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 369B 0.0s
=> [internal] load metadata for docker.io/library/python:3.12 2.7s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/4] FROM docker.io/library/python:3.12@sha256:872565c5ac89cafbab19419c699d80bda96e9d0f47a4790e5 60.1s
=> => resolve docker.io/library/python:3.12@sha256:872565c5ac89cafbab19419c699d80bda96e9d0f47a4790e52 0.1s
=> => sha256:872565c5ac89cafbab19419c699d80bda96e9d0f47a4790e5229bd3aeceb5da9 10.32kB / 10.32kB 0.0s
=> => sha256:db516ed76e844aa49c8ea4d15d5266de9b851f111c152b5f0ee6952f97095b3a 2.32kB / 2.32kB 0.0s
=> => sha256:d852c0dcc6f2a3022a0b05f8ee866b12c21b28bc5c9ce6ef40c29f57824a99ab 6.42kB / 6.42kB 0.0s
=> => sha256:795dbedde24d2c72dafd2b71fe36643552e56859c0e29c9db095ed54b825fbaa2 49.28MB / 49.28MB 5.4s
=> => sha256:89d573bf42b377ce6a5a0451c15388849686fa4058efd68999f3b014daeb5b55 25.62MB / 25.62MB 3.2s
=> => sha256:26dfe2fac1c486e9aaf41d1028ed30be2c442aa84af44462bc7bac8c148ffb13 67.78MB / 67.78MB 7.8s
=> => sha256:79d5bd8a8d262418bf22e705535ce38c6789dc72e319d76b30aafa5c331b6924 235.93MB / 235.93MB 17.3s
=> => sha256:cbd3b39d8e8e1aleff1429cc124cb11d4330928920457d6d801dc933c0342e8e 6.09MB / 6.09MB 6.6s
=> => extracting sha256:795dbedde24d2c72dafd2b71fe36643552e56859c0e29c9db095ed54b825fbaa2 10.6s
=> => sha256:ce2693a99e9219b3c7733afb165cf57a7ae8abe050af3c3625934ba3af876161 25.67MB / 25.67MB 9.5s
=> => sha256:4925315fece21039e064de701dd460e7e86b05edaa58bb4155aee62a3b334b3a 249B / 249B 8.2s
=> => extracting sha256:89d573bf42b377ce6a5a0451c15388849686fa4058efd68999f3b014daeb5b55 2.4s
=> => extracting sha256:26dfe2fac1c486e9aaf41d1028ed30be2c442aa84af44462bc7bac8c148ffb13 9.4s
=> => extracting sha256:79d5bd8a8d262418bf22e705535ce38c6789dc72e319d76b30aafa5c331b6924 26.8s
=> => extracting sha256:cbd3b39d8e8e1aleff1429cc124cb11d4330928920457d6d801dc933c0342e8e 0.8s
=> => extracting sha256:ce2693a99e9219b3c7733afb165cf57a7ae8abe050af3c3625934ba3af876161 2.4s
=> => extracting sha256:4925315fece21039e064de701dd460e7e86b05edaa58bb4155aee62a3b334b3a 0.0s
=> [internal] load build context 0.1s
=> => transferring context: 661B 0.0s
=> [2/4] WORKDIR /app 0.5s
=> [3/4] COPY . /app 0.1s
=> [4/4] RUN pip install --no-cache-dir -r requirements.txt 4.8s
=> => exporting to image 0.3s
=> => exporting layers 0.2s
=> => writing image sha256:ec595c2db85d1454752cad5a048670bba84617eee92fe8671eea0e9e45573d98 0.0s
=> => naming to docker.io/library/python-app 0.0s
○ user@deb:~/Documents/lab1$ docker run -p 9000:5000 python-app
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
* Restarting with stat
```

```

* Running on http://172.17.0.2:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 520-438-377
172.17.0.1 - - [27/Oct/2025 10:31:47] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [27/Oct/2025 10:31:48] "GET /favicon.ico HTTP/1.1" 404 -

```

For that previous step, the container must be running to be able to export it.
If it isn't, trying to refresh the web page doesn't work. Page is not found (error 404).



(When the docker is running)

```

user@deb:~/Documents/lab1$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
7a78b29072b2   python-app "python app.py"         3 minutes ago Up 3 minutes   0.0.0.0:9000->5000/tcp, [::]:9000->5000/tcp   youthful_bartik
user@deb:~/Documents/lab1$ docker export -o python-app.tar youthful_bartik
user@deb:~/Documents/lab1$ mkdir export-analyse
user@deb:~/Documents/lab1$ tar -xf python-app.tar -C export-analyse
user@deb:~/Documents/lab1$ ls
app.py  Dockerfile  export-analyse  python-app.tar  requirements.txt
user@deb:~/Documents/lab1$ cd export-analyse
ls
app  bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
user@deb:~/Documents/lab1/export-analyse$ ls etc
adduser.conf      default      group-      issue.net    magic.mime   pam.d        rc2.d        services    terminfo
alternatives      deluser.conf gshadow     kernel       mercurial    passwd       rc3.d        shadow      ucf.conf
apt               dpkg        gshadow-    ld.so.cache  mime.types   passwd-      rc4.d        shells      update-motd.d
bash.bashrc       emacs       gss         ld.so.conf   motd         perl         rc5.d        skel        wgetrc
bash_completion.d environment  host.conf   ld.so.conf.d mtab         profile     rc6.d        ssh         X11
bindresport.blacklist ethertypes  hostname    libaudit.conf mysql         profile.d   rcS.d        ssl         xattr.conf
ca-certificates   fonts       hosts       localtime    networks     protocols   resolv.conf  subgid
ca-certificates.conf fstab       ImageMagick-7 logcheck     nsswitch.conf python3      rmt         subuid
cron.daily         gai.conf    init.d      login.defs   opt          python3.13  rpc          subversion
debconf.conf      gprofng.rc inputrc     logrotate.d  os-release   rc0.d        security     sysctl.d
debian_version    group       issue      magic        pam.conf     rc1.d        selinux      systemd
user@deb:~/Documents/lab1/export-analyse$ ls app
app.py  Dockerfile  requirements.txt
user@deb:~/Documents/lab1/export-analyse$

```

Then we can cleanup the docker youthful_bartik :
docker rm youthful_bartik -force

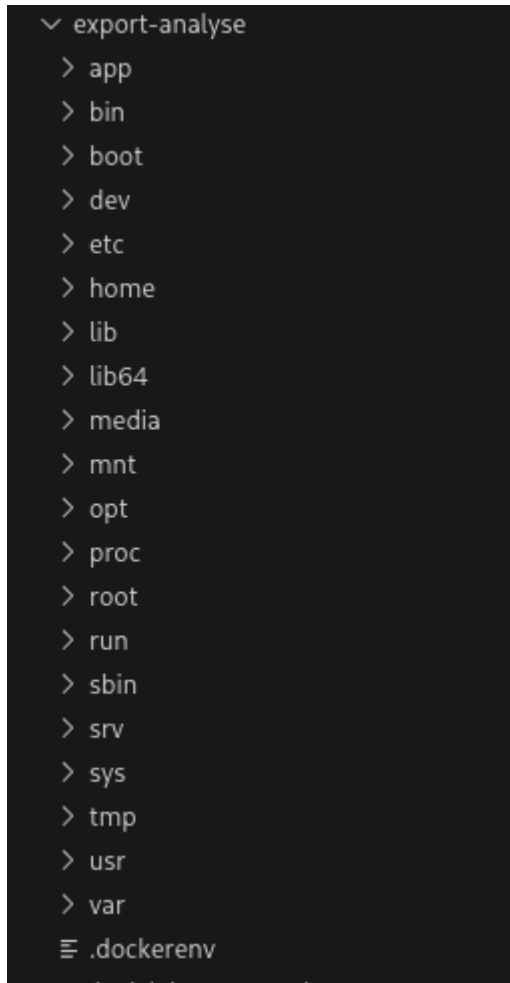
Nothing appears with “docker ps -a” :

```

user@deb:~/Documents/lab1$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
user@deb:~/Documents/lab1$ docker rm python-app -force

```

The structure of the archive :



This archive contains only the container's filesystem (its internal files).

We can inspect the Python environment (`/usr/local/lib`), application files (`/app`), configuration files (`/etc`), system packages installed, binaries and temporary files.

But it doesn't include the container metadata (name, ports, environment variables), the network configuration, the Docker image layers and the container's history.

Part 3 :

The IP address of a container attached to a Macvlan network is an address from the same subnet as the host's physical network (the LAN), assigned by Docker but based on the configuration specified when creating the Macvlan network.

```
user@deb:~/Documents/lab1/export-analyse$ docker container run -d --name app1 --network bridge nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
38513bd72563: Pull complete

10d18f46ee87: Pull complete

a8d825a0683a: Pull complete

a131bc1d4bd5: Pull complete

3818929ac19f: Pull complete

1498b1cfd15: Pull complete

c50c84d0ed4d: Pull complete

Digest: sha256:029d4461bd98f124e531380505ceea2072418fdf28752aa73b7b273ba3048903
Status: Downloaded newer image for nginx:latest
2d92363cb6ca13c460d6a3a0afe4895de147af916827fde0cbce622e502e6e65
user@deb:~/Documents/lab1/export-analyse$ docker container run -d --name app2 --network bridge nginx
eb42eff08337e51d78fa482bb275d86356f3ac1c626511aeafd2e15c393908c2
user@deb:~/Documents/lab1/export-analyse$ docker container ls
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
eb42eff08337	nginx	"/docker-entrypoint..."	About a minute ago	Up About a minute	80/tcp	app2
2d92363cb6ca	nginx	"/docker-entrypoint..."	About a minute ago	Up About a minute	80/tcp	app1

```
user@deb:~/Documents/lab1/export-analyse$ docker network inspect bridge
```

```
[
  {
    "Name": "bridge",
    "Id": "def8ef10c80ef624a4723bc4ec9bbec726e7a66bbad5418b314bada387943fb8",
    "Created": "2025-10-27T10:56:42.879496225+01:00",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": [
        {
          "Subnet": "172.17.0.0/16",
          "Gateway": "172.17.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "2d92363cb6ca13c460d6a3a0afe4895de147af916827fde0cbce622e502e6e65": {
        "Name": "app1",
        "EndpointID": "ff2ab6c579f5690c489103400bd981ceb7e5b06ad47e62ec6a66e9759de00e66",
        "MacAddress": "aa:50:d8:9c:fa:9f",
        "IPv4Address": "172.17.0.2/16",
        "IPv6Address": ""
      },
      "eb42eff08337e51d78fa482bb275d86356f3ac1c626511aeafd2e15c393908c2": {
        "Name": "app2",
        "EndpointID": "e99b517c641e638a1bde2704dfc3dc6d3368e7e42250e6ee3359392bcce98817",
        "MacAddress": "12:a8:34:17:95:f9",
        "IPv4Address": "172.17.0.3/16",
        "IPv6Address": ""
      }
    },
    "Options": {
      "com.docker.network.bridge.default_bridge": "true",
      "com.docker.network.bridge.enable_icc": "true",
      "com.docker.network.bridge.enable_ip_masquerade": "true",
      "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
      "com.docker.network.bridge.name": "docker0",
      "com.docker.network.driver.mtu": "1500"
    },
    "Labels": {}
  }
]
```

"Gateway": "172.17.0.1"


```

user@deb:~/Documents/lab1/export-analyse$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:43:b7:a6 brd ff:ff:ff:ff:ff:ff
    altname enx08002743b7a6
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 56997sec preferred_lft 56997sec
    inet6 fd17:625c:f037:2:e0d7:96a1:de6c:9822/64 scope global temporary dynamic
        valid_lft 85974sec preferred_lft 13974sec
    inet6 fd17:625c:f037:2:5545:69c4:47a7:e86c/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85974sec preferred_lft 13974sec
    inet6 fd17:625c:f037:2:c209:70f2:bf57:6fcb/64 scope global temporary dynamic
        valid_lft 85974sec preferred_lft 13974sec
    inet6 fd17:625c:f037:2:a00:27ff:fe43:b7a6/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 85974sec preferred_lft 13974sec
    inet6 fe80::313f:d389:a254:a421/64 scope link
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe43:b7a6/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 06:84:e5:1d:a9:4e brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever

```

```

user@deb:~/Documents/lab1/export-analyse$ docker exec -it host-container1 hostname -I
10.0.2.100 172.17.0.1
user@deb:~/Documents/lab1/export-analyse$ docker exec -it host-container2 hostname -I
hostname: unrecognized option: I
BusyBox v1.37.0 (2025-08-05 16:40:33 UTC) multi-call binary.

Usage: hostname [-sidf] [HOSTNAME | -F FILE]

Show or set hostname or DNS domain name

        -s          Short
        -i          Addresses for the hostname
        -d          DNS domain name
        -f          Fully qualified domain name
        -F FILE     Use FILE's content as hostname
user@deb:~/Documents/lab1/export-analyse$ docker exec -it host-container2 ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: tap0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 65520 qdisc fq_codel state UP qlen 1000
    link/ether 96:f9:29:e4:7b:c2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.100/24 scope global tap0
        valid_lft forever preferred_lft forever
    inet6 fe80::94f9:29ff:fee4:7bc2/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP
    link/ether 7a:d7:fa:13:8b:25 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::78d7:faff:fe13:8b25/64 scope link
        valid_lft forever preferred_lft forever
16: veth9e44988@tap0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP
    link/ether 76:d5:3f:81:be:b5 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::74d5:3fff:fe81:beb5/64 scope link
        valid_lft forever preferred_lft forever
17: veth0e3912f@tap0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP
    link/ether 96:e0:f9:ff:5c:66 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::94e0:f9ff:feff:5c66/64 scope link
        valid_lft forever preferred_lft forever

```

So, both host-container1 and host-container2 use the same IP address as the host : 10.0.2.100.

Regarding open ports :

Ports are shared with the host, not isolated per container.

The host-container1 (Nginx) listen on port 80 of the host machine. This was confirmed by running curl http://localhost, which returned the Nginx welcome page :

```
root@deb:/# curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

The host-container2 (Alpine sleep infinite) doesn't listen on any port.

```

user@deb:~/Documents/lab1/export-analyse$ docker container run -d --name host-container3 --network host nginx
520906ddde5237cbe7904b617dclac793b9d56356c2b2a6c6414d4093e9435e
user@deb:~/Documents/lab1/export-analyse$ docker logs host-container3
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/10/27 19:18:13 [emerg] 1#1: bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
2025/10/27 19:18:13 [emerg] 1#1: bind() to [::]:80 failed (98: Address already in use)
nginx: [emerg] bind() to [::]:80 failed (98: Address already in use)
2025/10/27 19:18:13 [notice] 1#1: try again to bind() after 500ms
2025/10/27 19:18:13 [emerg] 1#1: bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
2025/10/27 19:18:13 [emerg] 1#1: bind() to [::]:80 failed (98: Address already in use)
nginx: [emerg] bind() to [::]:80 failed (98: Address already in use)
2025/10/27 19:18:13 [notice] 1#1: try again to bind() after 500ms
2025/10/27 19:18:13 [emerg] 1#1: bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
2025/10/27 19:18:13 [emerg] 1#1: bind() to [::]:80 failed (98: Address already in use)
nginx: [emerg] bind() to [::]:80 failed (98: Address already in use)
2025/10/27 19:18:13 [notice] 1#1: try again to bind() after 500ms
2025/10/27 19:18:13 [emerg] 1#1: bind() to 0.0.0.0:80 failed (98: Address already in use)
nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
2025/10/27 19:18:13 [emerg] 1#1: bind() to [::]:80 failed (98: Address already in use)
nginx: [emerg] bind() to [::]:80 failed (98: Address already in use)
2025/10/27 19:18:13 [notice] 1#1: try again to bind() after 500ms
2025/10/27 19:18:13 [emerg] 1#1: still could not bind()
nginx: [emerg] still could not bind()
user@deb:~/Documents/lab1/export-analyse$

```

In this current set up, host-container1 is already running Nginx on port 80 in host mode.

As explained before, host mode doesn't isolate ports → every container shares the host's network stack.

Two processes cannot listen on the same port of the same IP.

Docker has started the container but Nginx inside host-container3 has fail to start, see above.

```

user@deb:~/Documents/lab1/export-analyse$ docker container run -d --name none-container2 --network none nginx
bb89fdb2f92674524bc4e2a33f53db288afd0a6f0ac60f084ccbd2eb1f128c2f
user@deb:~/Documents/lab1/export-analyse$ docker container run -d --name none-container1 --network none nginx
17c379bdb7312228dde061b0043a1bf7fa7a1ad42b85f323cb26d4f7fc8d5d0b
user@deb:~/Documents/lab1/export-analyse$ docker container ls
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
17c379bdb731   nginx     "/docker-entrypoint.    About a minute Up About a minute           none-container1
bb89fdb2f926   nginx     "/docker-entrypoint.    2 minutes ago Up 2 minutes           none-container2
7d9f8e14155a   alpine    "sleep infinite"        16 hours ago  Up 16 hours           host-container2
5671f47a9404   nginx     "/docker-entrypoint.    16 hours ago  Up 16 hours           host-container1
eb42eff08337   nginx     "/docker-entrypoint.    17 hours ago  Up 17 hours           app2
2d92363cb6ca   nginx     "/docker-entrypoint.    17 hours ago  Up 17 hours           app1
user@deb:~/Documents/lab1/export-analyse$ docker network inspect none
[
  {
    "Name": "none",
    "Id": "7e33fa41fb5dbbe76c1e61e70bfafc36039dc7d703a9f68e5f146c529cf12931",
    "Created": "2025-10-23T19:42:32.118162536+02:00",
    "Scope": "local",
    "Driver": "null",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": null,
      "Config": null
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "17c379bdb7312228dde061b0043a1bf7fa7a1ad42b85f323cb26d4f7fc8d5d0b": {
        "Name": "none-container1",
        "EndpointID": "30c2a7003dd833a877edde468abbfa7090fe810f0609bb3a554765cf7e5e97c3",
        "MacAddress": "",
        "IPv4Address": "",
        "IPv6Address": ""
      },
      "bb89fdb2f92674524bc4e2a33f53db288afd0a6f0ac60f084ccbd2eb1f128c2f": {
        "Name": "none-container2",
        "EndpointID": "c38e4da8ea11f46b34a24e887140547db59cfc6716f77ffa564aa17744dc83e9",
        "MacAddress": "",
        "IPv4Address": "",
        "IPv6Address": ""
      }
    },
    "Options": {},
    "Labels": {}
  }
]

```

Both containers cannot access the internet and cannot communicate with each other or with the host.

```

user@deb:~/Documents/lab1/export-analyse$ docker exec -it first ifconfig
docker exec -it second ifconfig
eth0      Link encap:Ethernet  HWaddr F2:62:F3:57:6C:1E
          inet addr:172.17.0.4  Bcast:172.17.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:3 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:866 (866.0 B)  TX bytes:126 (126.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

eth0      Link encap:Ethernet  HWaddr F2:62:F3:57:6C:1E
          inet addr:172.17.0.4  Bcast:172.17.255.255  Mask:255.255.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:11 errors:0 dropped:0 overruns:0 frame:0
          TX packets:3 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:866 (866.0 B)  TX bytes:126 (126.0 B)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)

```

inet addr:172.17.0.4 proves that they are sharing the same network namespace.

They're sharing the same network interfaces lo and eth0.

checking connectivity between the containers :

```
user@deb:~/Documents/lab1/export-analyse$ docker exec -it first nc -nlp 1234
aa
This is written in the second container
This is written in the first container
```

We got the same screen in both terminals.

```
user@deb:~/Documents/lab1/export-analyse$ docker exec -it first hostname
docker exec -it second hostname
79dd79cc5739
79dd79cc5739
```

Both containers return the same hostname because they share the same network identity.

By running `docker rm first` :

the second container stops and becomes unusable. It depends on the first container's network namespace which is deleted at the same time.

When we want tight coupling between containers, that must share network resources, like a sidecar container, we can deploy two containers in the same network namespace.

If both first and second listen to the same port, a port conflict occurs by trying to bind the second container to that port. The second process fail to bind to the port already in use. (Only one can listen on a given port at a time)